Notes: Colliard, Foucault, and Hoffmann (JF, 2021)

Inventory Management, Dealers' Connections, and Prices in Over-the-Counter Markets

Contents

1 Big picture

- **Question.** How do **dealer inventories** and **heterogeneous trading connections** jointly determine prices, bid-ask spreads, and efficiency in OTC markets?
- **Setting.** The paper builds a stylized OTC market with three groups:
  - **core dealers**, who are tightly connected and trade competitively in a centralized core market,
  - **connected peripheral dealers**, who can bargain with other peripheral dealers and also unload inventory in the core market,
  - **unconnected peripheral dealers**, who can only bargain bilaterally in the periphery.
- **Core challenge.** Standard inventory models capture how dealers shade prices when they are long or short, but they usually abstract from the network structure of the interdealer market. Pure network models, in contrast, typically do not focus on the role of inventory imbalances. This paper's key contribution is to make these two frictions interact.
- **Main mechanism.** Peripheral dealers derive market power from two sources:
  1. **connectedness**: connected dealers have a better outside option because they can trade with core dealers;
  2. **uncrowdedness**: dealers on the scarce side of the peripheral market have more bargaining power.

  The paper's most novel result is that **connectedness affects prices if and only if peripheral dealers' aggregate inventory is not zero**. If the periphery is balanced, all peripheral trades occur at the core price even when some dealers are better connected.
- **Why it matters.** This gives a new way to think about liquidity in bond, currency, and other OTC markets. Whether central dealers charge tighter or wider prices than peripheral dealers depends not just on network centrality, but also on where inventory pressure sits in the system.
- **Main theoretical results.**
  - The **core-market price** falls when core dealers or connected peripheral dealers bring more net inventory into the core.
  - The interdealer market can settle into three regimes: **ACD** (active connected dealers), **ICS** (inactive connected sellers), or **ICB** (inactive connected buyers).
  - **Price dispersion** in the peripheral market rises with the size of the peripheral inventory imbalance.
  - Connected dealers trade at better prices than unconnected dealers, but the gap is much larger for dealers on the **crowded side** of the market.
  - Connected dealers give **more competitive quotes to clients** than unconnected dealers because they can unwind inventory more favorably.
  - Equilibrium trading among peripheral dealers is generally **inefficient**; decentralized bargaining leaves too much inventory risk stranded on some dealers.
- **Main empirical/simulation message.** If an empiricist regresses OTC prices on dealer connectedness alone, the estimates can be misleading. The effect of connectedness must be interacted with:

  - whether the dealer is buying or selling,
  - whether that side of the market is crowded,
  - the size of dealers' aggregate inventory imbalance.

- **Economic takeaway.** OTC price formation is not just about who has inventory. It is about **who has inventory and who has outside options**. Network position matters most precisely when inventory imbalances make outside options valuable.

2 Data and measurement (key objects)

2.1 Market participants and network structure

- **Clients.** End-investors do not trade directly with one another. Each client is matched with exactly one dealer.
- **Core dealers.** These dealers are highly interconnected and can be thought of as operating in a frictionless interdealer segment. They absorb inventory from connected peripheral dealers in period 2.
- **Peripheral dealers.** These are smaller intermediaries who bargain bilaterally.
  - A fraction $\lambda \leq 1/2$ are **unconnected** and cannot access the core market.
  - The remaining fraction $1 - \lambda$ are **connected** and can trade with core dealers if peripheral bargaining fails.
- **Why this distinction matters.** Connectedness is the paper's central network friction. It gives one class of peripheral dealers an outside option that others do not have.

2.2 Timing and trading process

- **Three periods.**
  1. **Period 1:** dealers trade with clients and peripheral dealers bargain bilaterally.
  2. **Period 2:** connected peripheral dealers and core dealers trade in the core market.
  3. **Period 3:** the asset payoff is realized and inventory holding costs are paid.
- **Client side.** Each client has private valuation $\Delta \in \{-L, L\}$. The assumption $L > \max(C_s, C_b)$ ensures gains from trade with dealers always exist.
- **Peripheral bargaining protocol.** Peripheral dealers arrive sequentially. Dealer τ makes a take-it-or-leave-it offer to dealer $\tau + 1$. If rejected:
  - the rejected connected dealer can later trade in the core market,
  - the rejected unconnected dealer must continue in the peripheral chain or carry inventory.
- **Random market length.** The number of peripheral subperiods is random. Each subperiod is the last with probability $1 - \psi$.

2.3 Inventory and cost objects

- **Dealer positions.** After trading with a client, dealer i holds $z_{i0} = +1$ if he bought from the client and $z_{i0} = -1$ if he sold to the client.
- **Inventory holding costs.**
  - C_s: per-unit cost of carrying a long position.
  - C_b: per-unit cost of carrying a short position.

  These costs capture risk-bearing or overnight funding costs in reduced form.
- **Aggregate inventories.**

$$z_0^{pe} = 2\alpha^{pe} - 1, \qquad z_0^{co} = \kappa(2\alpha^{co} - 1).$$

  Here:
  - α^{pe} is the mass of peripheral dealers with a long position at date 0,
  - α^{co} is the mass of core dealers with a long position at date 0,
  - κ controls the size of the core sector relative to the peripheral sector.
- **Core inventory shock.** Core dealers also receive an end-of-day shock ϵ_{i3} with distribution $\Pr(\epsilon_{i3} \leq x) = \Gamma(x/\kappa)$. This makes expected inventory costs strictly convex and gives a well-behaved downward-sloping demand curve in the core market.

2.4 Simulation design and empirical counterparts

- **What the paper does empirically.** This is not a structural estimation on market data. Instead, the paper simulates **250 trading days** from the model.
- **Simulation design.**
  - Each day draws new values of α^{pe} and α^{co}.
  - Other parameters are fixed across days.
  - The baseline calibration sets $\lambda = 0.4$.
- **Recorded objects.**
  - each transaction price,
  - seller type and buyer type,
  - daily peripheral and core aggregate inventories,
  - whether dealers end the day with inventory.
- **Why this matters.** The simulations show what an empiricist would recover from regression-style evidence if the true data-generating process were the paper's model.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dealer payoff and inventory dynamics

The final payoff of dealer i is

$$\Pi_{i3}(z_{i3}, m_{i3}) = vz_{i3} - C(z_{i3}) + m_{i3}, \tag{1}$$

with inventory cost function

$$C(x) = \begin{cases} C_s x, & x > 0, \\ C_b |x|, & x < 0. \end{cases}$$

The dealer's final inventory evolves according to

$$z_{i3} = z_{i0} + q_{i1} + q_{i2} + \epsilon_{i3}, \tag{2}$$

and final cash holdings are

$$m_{i3} = m_{i0} - p_{i1}q_{i1} - p_{i2}q_{i2}. \tag{3}$$

Interpretation.

- The model is all about how dealers try to get from $z_{i0} \in \{-1, +1\}$ to something closer to zero before costly inventory is carried overnight.
- Trading with clients creates the inventory problem; interdealer trading solves it only imperfectly.

3.2 Core-market equilibrium

A core dealer chooses period-2 demand q_{i2}^{co*} to maximize expected payoff net of expected inventory cost. The expected inventory cost is

$$\bar{C}(q_{i2}; z_{i0}) = C_s \int_{-(z_{i0}+q_{i2})}^{+\infty} (z_{i0} + q_{i2} + \epsilon)\, d\Gamma(\kappa^{-1}\epsilon) - C_b \int_{-\infty}^{-(z_{i0}+q_{i2})} (z_{i0} + q_{i2} + \epsilon)\, d\Gamma(\kappa^{-1}\epsilon). \tag{7}$$

The core dealer's first-order condition is

$$p^{co} + \bar{C}'(q_{i2}^{co*}; z_{i0}) = 0. \tag{8}$$

Solving yields individual core demand

$$q_{i2}^{co*}(p^{co}, z_{i0}) = -z_{i0} - \kappa\Gamma^{-1}\left(\frac{p^{co} + C_s}{C_s + C_b}\right), \qquad p^{co} \in (-C_s, C_b), \tag{9}$$

and average core demand

$$q_2^{co*}(p^{co}, z_0^{co}) = -z_0^{co} - \kappa\Gamma^{-1}\left(\frac{p^{co} + C_s}{C_s + C_b}\right). \tag{10}$$

Connected peripheral dealers bring order flow $-\Sigma$ into the core market, so market clearing implies

$$q_2^{co*}(p^{co*}, z_0^{co}) = \Sigma. \tag{12}$$

Hence the equilibrium core price is

$$p^{co*} = \Gamma(-\kappa^{-1}z^*)C_b - \big(1 - \Gamma(-\kappa^{-1}z^*)\big)C_s, \qquad z^* = z_0^{co} + \Sigma. \tag{13}$$

Interpretation.

- The core market price is the model's reference price.
- It falls when the core sector has to absorb more net inventory, either because core dealers themselves are long or because connected peripheral dealers dump inventory on them.
- In the thick-core case ($\kappa \to \infty$), the core market behaves like a deep outside option for connected dealers.

3.3 Peripheral bargaining and continuation values

An unconnected buyer and seller solve

$$V_b^U = \max_{p_b \in (-C_s, C_b)} \Big\{-\phi_b(p_b)p_b - [1 - \phi_b(p_b)]C_b\Big\}, \tag{14}$$

$$V_s^U = \max_{p_s \in (-C_s, C_b)} \Big\{\phi_s(p_s)p_s - [1 - \phi_s(p_s)]C_s\Big\}. \tag{15}$$

A connected buyer and seller solve

$$V_b^C = \max_{p_b \in (-C_s, C_b)} \Big\{-\phi_b(p_b)p_b - [1 - \phi_b(p_b)]p^{co*}\Big\}, \tag{16}$$

$$V_s^C = \max_{p_s \in (-C_s, C_b)} \Big\{\phi_s(p_s)p_s + [1 - \phi_s(p_s)]p^{co*}\Big\}. \tag{17}$$

Acceptance probabilities take the cutoff form

$$\phi_b(p_b) = \begin{cases} 0, & p_b < V_s^U, \\ \lambda\pi_s, & p_b \in [V_s^U, V_s^C), \\ \pi_s, & p_b \geq V_s^C, \end{cases} \qquad \phi_s(p_s) = \begin{cases} 0, & p_s > -V_b^U, \\ \lambda\pi_b, & p_s \in (-V_b^C, -V_b^U], \\ \pi_b, & p_s \leq -V_b^C. \end{cases} \tag{18}$$

Interpretation.

- Connected dealers have weakly better continuation values because rejection still leaves access to the core market.
- This is the paper's central source of bargaining asymmetry.
- Whether that asymmetry actually moves prices depends on whether one side of the peripheral market is crowded.

3.4 Equilibrium regimes in the peripheral market

The peripheral market admits three economically important pure-strategy regimes:

- **ACD (active connected dealers):** connected buyers and sellers both remain active in peripheral bargaining.
- **ICS (inactive connected sellers):** connected sellers prefer to go directly to the core market.
- **ICB (inactive connected buyers):** connected buyers prefer to go directly to the core market.

Proposition 1 (economic content).

- If the periphery is **long** overall ($\alpha^{pe} > 1/2$) and the core is sufficiently **short**, connected sellers become inactive and the market moves toward ICS.
- If the periphery is **short** overall ($\alpha^{pe} < 1/2$) and the core is sufficiently **long**, connected buyers become inactive and the market moves toward ICB.
- Otherwise, the market typically stays in the ACD region.
- With a non-thick core, mixed-strategy regions can appear between these pure-strategy zones.

Interpretation.

- Which regime obtains depends on how attractive the core-market outside option is relative to bargaining in the periphery.
- Extreme and opposite-signed inventory imbalances between core and periphery make direct access to the core market especially valuable for one connected side.

3.5 Equilibrium prices in the peripheral market

Define

$$\rho_k^C = \frac{1-\pi_k}{1-\pi_b\pi_s\lambda}, \qquad \rho_k^U = -\lambda\pi_k\rho_{-k}^C, \qquad k \in \{b, s\}.$$

Then Proposition 2 gives the transaction prices among peripheral dealers:

ACD regime

$$p_s^U = p^{co*} + \rho_s^U(C_s + p^{co*}), \qquad p_s^C = p^{co*} + \rho_s^C(C_b - p^{co*}), \tag{21}$$

$$p_b^U = p^{co*} - \rho_b^U(C_b - p^{co*}), \qquad p_b^C = p^{co*} - \rho_b^C(C_s + p^{co*}). \tag{22}$$

ICS regime

$$p_s^U = p^{co*} + \rho_s^U(C_s + p^{co*}), \qquad p_s^C \text{ is not observed}, \tag{23}$$

$$p_b^U = p^{co*} - \rho_b^C(C_s + p^{co*}), \qquad p_b^C = p^{co*} - \rho_b^C(C_s + p^{co*}). \tag{24}$$

ICB regime

$$p_s^U = p^{co*} + \rho_s^C(C_b - p^{co*}), \qquad p_s^C = p^{co*} + \rho_s^C(C_b - p^{co*}), \tag{25}$$

$$p_b^U = p^{co*} - \rho_b^U(C_b - p^{co*}), \qquad p_b^C \text{ is not observed}. \tag{26}$$

The paper defines market power as

$$M_b^i = \frac{C_b + V_b^i}{C_b - p^{co*}}, \tag{27}$$

$$M_s^i = \frac{V_s^i + C_s}{p^{co*} + C_s}. \tag{28}$$

Interpretation.

- Connected dealers receive the largest price concessions whenever they are observed in the peripheral market.
- But connectedness alone is *not* enough to generate price dispersion. If $\lambda = 0$ or $z_0^{pe} = 0$, all peripheral trades collapse to the core price.
- The most powerful dealers are those who are both **connected** and on the **uncrowded side** of the market.

3.6 Dealer-to-client quotes

Let P_k^i be the expected price at which a dealer of type (k, i) unwinds his inventory in the interdealer market. Then dealer-client quotes are

$$\text{bid}^i = P_s^i - (1-\beta)(P_s^i + L), \tag{29}$$

$$\text{ask}^i = P_b^i + (1-\beta)(L - P_b^i). \tag{30}$$

Interpretation.

- Dealers pass through some of their interdealer bargaining advantage to customers.
- Connected dealers can post tighter quotes because they expect to unwind positions at better prices.
- This creates cross-dealer price dispersion in the D2C market even though clients themselves are homogeneous.

3.7 Trading costs and welfare

Total trading costs borne by peripheral dealers' customers are

$$TC^{pe}(\Xi) = (1-\alpha^{pe})(\lambda\,\text{ask}^U + (1-\lambda)\,\text{ask}^C) - \alpha^{pe}(\lambda\,\text{bid}^U + (1-\lambda)\,\text{bid}^C). \tag{34}$$

The paper rewrites this as

$$TC^{pe}(\Xi) = (1-\beta)L + \beta IC(\Xi) - \beta W(\Xi), \tag{35}$$

where:

- $IC(\Xi)$ is aggregate inventory cost without peripheral rebalancing,
- $W(\Xi)$ is the gains from trade realized among peripheral dealers.

Interpretation.

- Better interdealer reallocation reduces customer trading costs directly.
- Frictions in the dealer network are therefore not just a dealer-side problem; they are passed through to end-investors.
- Proposition 4 shows the decentralized equilibrium is inefficient when $\alpha^{pe} \neq 1/2$: dealers engage in rent-seeking behavior rather than minimizing total inventory cost.

3.8 Simulation regressions used in the paper

For interdealer transactions, the paper simulates the regression

$$\hat{p}_{n,d} = \alpha + \sum_{(i,j)\in\{C,U,\mathrm{co}\},\,(i,j)\neq(\mathrm{co},\mathrm{co})} \beta^{i,j} D^{j}_{i,n,d} + \varepsilon_{n,d}, \tag{32}$$

where $\hat{p}_{n,d} = p_{n,d} - p^{co}_{d}$ is the transaction price relative to the core price on day d.

For client transactions, it simulates

$$\hat{p}^{cu}_{n,d} = \alpha + \sum_{i\in\{C,U\}} \beta^{cu,i} D^{cu,i}_{n,d} + \sum_{i\in\{C,U\}} \beta^{i,cu} D^{i,cu}_{n,d} + \varepsilon_{n,d}, \tag{33}$$

where the dependent variable is the client-transaction price relative to the average price that core dealers offer that day.

Interpretation.

- These are not estimation equations from real data; they are diagnostic regressions run on simulated data.
- Their role is to show what patterns an empiricist should look for in actual OTC data.
- The key lesson is that sorting or interacting by z^{pe}_0 is essential. Pooling all days can wash out the true structure of connectedness effects.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard OTC regressions miss

- **Connectedness as a single main effect is not enough.** The same dealer can benefit a lot from connectedness on one day and much less on another, depending on whether his side of the market is crowded.
- **Inventory alone is not enough.** Aggregate dealer inventories matter differently for connected and unconnected dealers because only the former can escape to the core market.
- **Average centrality discounts/premia can be misleading.** A regression that ignores the sign and size of aggregate inventory pressure can mix together opposite effects and understate the true mechanism.

4.2 What identifies the mechanism here

The model is empirically sharp because it links a small set of observables to the economic mechanism:

1. **D2D price differentials** across connected and unconnected dealers with the same inventory sign;
2. **D2D price dispersion** relative to the core price;
3. **inventory carryover probabilities** for unconnected versus connected dealers;
4. **D2C quote differences** across dealer types;
5. **responses to core-inventory shocks**, which should flip sign depending on whether the periphery is crowded on the buy or sell side.

Interpretation. These are the theory-paper analog of identification moments. If the data show that connectedness only matters when $z_0^{pe} \neq 0$, and especially for dealers on the crowded side, that would be hard to reconcile with simpler models.

4.3 Why the auxiliary implications are useful

The paper's sharpest distinctive predictions are:

- if $\lambda = 0$ or $z_0^{pe} = 0$, all interdealer prices collapse to the core price;
- price dispersion rises with $|z_0^{pe}|$;
- a positive shock to core dealers' inventory increases or decreases dispersion depending on the sign of z_0^{pe};
- connected dealers' D2C spreads depend on inventory imbalances, whereas in standard inventory models inventories mainly move the midpoint, not the spread.

These are especially useful because they distinguish this model from pure inventory models and from pure network models.

4.4 Why the welfare comparison is meaningful

- The planner benchmark does *not* rely on superior information.
- The social planner is restricted to the same information that dealers have when making decisions.
- Therefore, the gap between equilibrium and first best is genuinely about **strategic bargaining inefficiency**, not about informational omniscience by the planner.

4.5 Relation to the literature

- **Versus standard inventory models (Ho–Stoll, Grossman–Miller):** this paper makes spreads and price dispersion depend on network position, not just inventory per se.
- **Versus network models of OTC trading:** this paper emphasizes inventory imbalances as the state variable that activates network effects.

- **Broader contribution:** it provides a unified explanation for why dealer centrality can sometimes look beneficial to clients, sometimes not, and why the answer depends on where inventory pressure sits in the dealer network.

5 Tables and figures

5.1 Figures 1 and 2: timeline and peripheral bargaining protocol

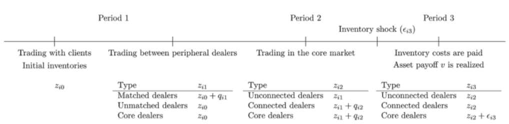

Figure 1. Timeline. This figure shows the evolution of dealer i's inventory position (z_{it}) in each period t, given his trade (q_{it}) in each period and his final inventory shock (ϵ_{i3}; set at zero for peripheral dealers).

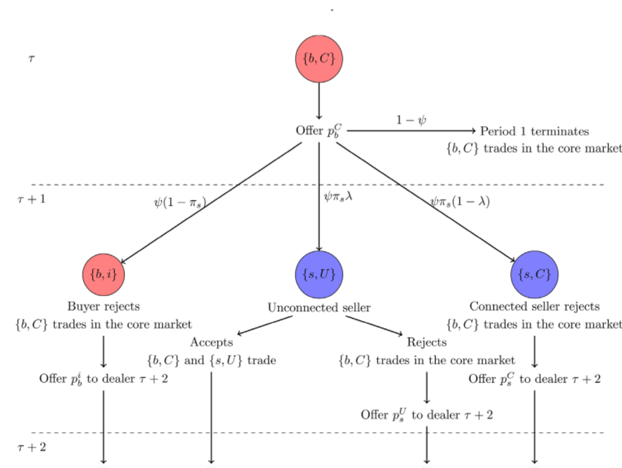

Figure 2. Trading in the periphery. This figure depicts the trading process in the peripheral market for a particular sequence of arrivals. Circles with a "b" designate buyers while circles with an "s" designate sellers. We denote by p_b^j the offer made by a buyer of type $j \in \{C, U\}$ and by p_s^j the offer made by a seller of type $j \in \{C, U\}$, where C designates a connected dealer and U designates an unconnected dealer. (Color figure can be viewed at wileyonlinelibrary.com)

Read:

- Figure 1 compresses the model into one picture: clients create inventory in period 1, peripheral dealers try to rebalance bilaterally, connected dealers can then trade in the core market, and inventory costs are paid only at the end.

- Figure 2 shows exactly where connectedness enters. A connected dealer whose negotiation fails can still trade in the core market; an unconnected dealer cannot. That asymmetry is the source of the continuation-value gap ($V^C - V^U$).

- The figure also shows that the bargaining game is sequential, not simultaneous. This is why the probability of meeting a counterpart on the other side matters so much.

5.2 Figures 3 and 4: equilibrium regimes

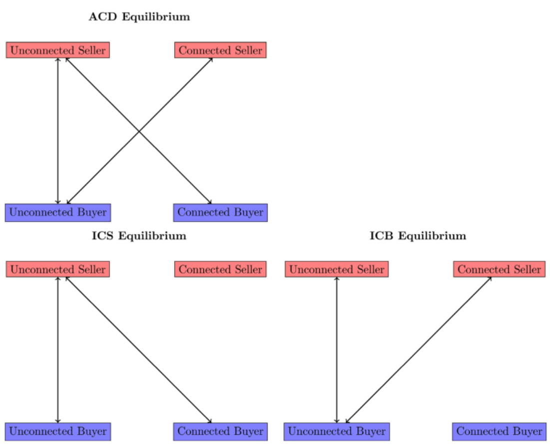

Figure 3. Equilibrium configurations. An arrow going from one type of trader to another indicates that the former makes offers to the latter in equilibrium. "ACD": active connected dealers; "ICS": inactive connected sellers; "ICB": inactive connected buyers. (Color figure can be viewed at wileyonlinelibrary.com)

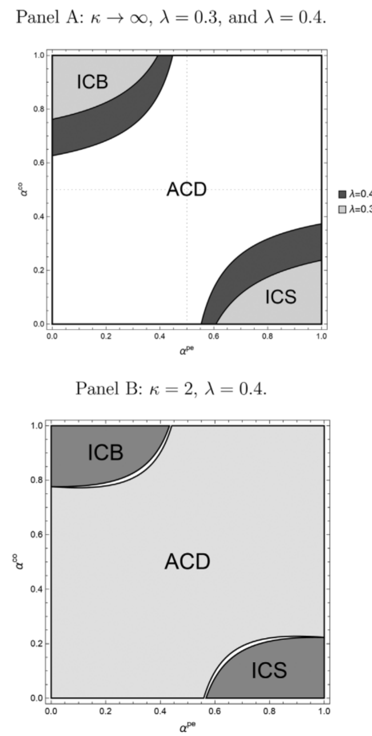

Figure 4. Equilibrium type as a function of α^{pe} and α^{co}. In Panel A, the figure depicts the equilibrium strategy profile (ACD: active connected dealers; ICB: inactive connected buyers; ICS: inactive connected sellers) in the thick core market case that arises for different values of α^{pe} and α^{co}, the masses of peripheral and core dealers with a long position at date 0. For illustration, we display the results for two different values of the mass of unconnected peripheral dealers, λ (0.3 and 0.4), where the additional area covered by the darker shades corresponds to a higher value of λ. In Panel B, the figure provides the corresponding equilibrium strategy profile obtained when the core market is not thick. The shaded areas represent the $(\alpha^{pe}, \alpha^{co})$ pairs for which an ACD, ICS, or ICB equilibrium is obtained, whereas a mixed equilibrium obtains in the remaining white areas.

Read:

- Figure 3 gives the clean graphical meaning of the three regimes:
  - in ACD, both connected buyers and connected sellers remain active in the peripheral market;
  - in ICS, connected sellers bypass the periphery and go straight to the core;
  - in ICB, connected buyers do the same.
- Figure 4 maps these regimes into the $(\alpha^{pe}, \alpha^{co})$ space.
  - The central region is ACD.
  - The upper-left region is ICB: the periphery is buy-crowded while the core is sell-crowded.
  - The lower-right region is ICS: the periphery is sell-crowded while the core is buy-crowded.
- The darker region for larger λ shows that a less connected periphery makes inactivity regions larger. In other words, more unconnected dealers make bargaining frictions more severe.

5.3 Figure 5: ordering of equilibrium prices

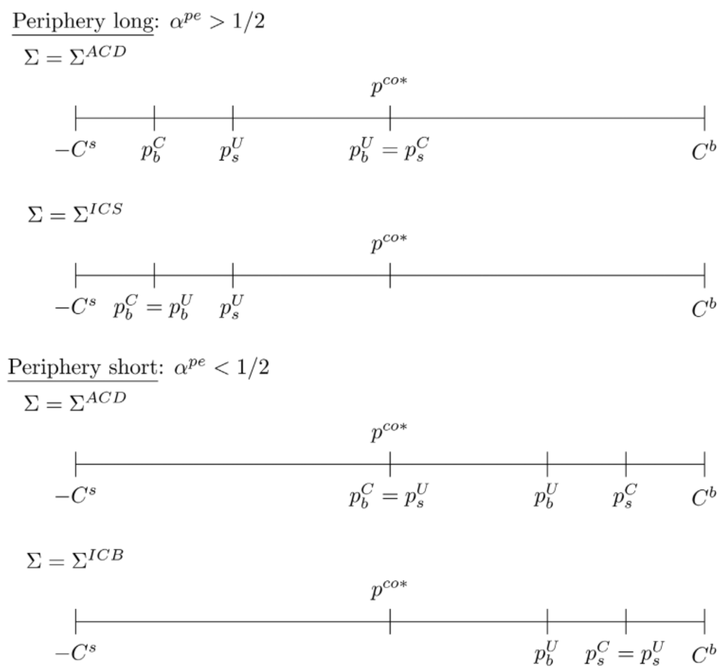

Figure 5. Equilibrium prices. This figure shows the position of equilibrium prices in the peripheral market relative to the equilibrium core market price (p^{co*}) in each possible equilibrium regime (ACD: active connected dealers; ICB: inactive connected buyers; ICS: inactive connected sellers) for each possible realization of the mass (α^{pe}) of peripheral dealers with a long position (see Proposition 2). We denote by p_b^j the offer made by a buyer of type $j \in \{C, U\}$ and by p_s^j the offer made by a seller of type $j \in \{C, U\}$, where C designates a connected dealer and U designates an unconnected dealer.

Read:

- Figure 5 is the key pricing diagram. It shows all peripheral prices relative to the core price p^{co*}.
- When observed, **connected sellers get the highest prices** and **connected buyers get the lowest prices**. They extract the largest concessions because they have the strongest outside options.
- The figure also makes the paper's headline point visually obvious: when the periphery is imbalanced, prices fan out around the core price; when it is balanced, they collapse back toward the core price.
- The ordering of p_b^U and p_s^U changes across regimes, which is why empirical work must condition on the sign of the aggregate peripheral inventory imbalance.

5.4 Tables I and II: simulated regressions for D2D and D2C prices

Table I

Dealer-to-Dealer (D2D) Prices and Dealer Types

This table presents estimates of regression (32). Column (1) gives the coefficients estimated on all D2D transactions. Columns (2) to (5) give the coefficients estimated on four subsamples, corresponding to the four quartiles of α^{pe}. t-statistics are given in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the trading day level. The constant, as well as the coefficients $\beta^{co,C}$ and $\beta^{C,co}$ are equal to zero and omitted. There are 1,175 trades between unconnected dealers, 4,914 trades between a connected and an unconnected dealer, and 10,036 trades between connected and core dealers.

	(1)	(2)	(3)	(4)	(5)
$\beta^{C,U}$	0.193***	0.460***	0.285***	0.010**	0.000
	(9.52)	(11.80)	(12.06)	(2.56)	(0.00)
$\beta^{U,C}$	−0.181***	0.000	−0.000***	−0.182***	−0.498***
	(−8.44)	(0.00)	(−20.71)	(−5.37)	(−17.97)
$\beta^{U,U}$	0.008	0.375***	0.148***	−0.083***	−0.342***
	(0.28)	(5.20)	(4.52)	(−4.85)	(−8.48)
# Obs.	17,375	4,161	4,402	4,674	4,138

Table II

Dealer-to-Customer (D2C) Prices and Dealer Types

This table presents estimates of regression (33). Column (1) gives the coefficients estimated on all D2C transactions. Columns (2) to (5) give the coefficients estimated on four subsamples, corresponding to the four quartiles of α^{pe}. t-statistics are given in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively. Standard errors are clustered at the trading day level. The constant and the coefficient $\beta^{cu,co}$ are equal to zero and omitted. There are 9,849 trades between clients and unconnected dealers and 14,950 trades between clients and connected dealers.

	(1)	(2)	(3)	(4)	(5)
$\beta^{cu,U}$	−0.159***	0.056***	0.020***	−0.129***	−0.368***
	(−9.00)	(7.91)	(8.26)	(−5.53)	(−22.19)
$\beta^{U,cu}$	0.176***	0.364***	0.190***	−0.006	−0.054***
	(9.26)	(14.09)	(11.90)	(−1.56)	(−12.74)
$\beta^{cu,C}$	0.034***	0.140***	0.078***	0.002**	0.000
	(7.44)	(12.39)	(12.00)	(2.32)	(0.00)
$\beta^{C,cu}$	−0.029***	−0.000	0.000	−0.049***	−0.144***
	(−7.17)	(.)	(.)	(−5.74)	(−20.54)
# Obs.	34,738	8,833	8,450	8,817	8,638

Read:

- **Table I (D2D prices).** On pooled days, connected sellers get better prices than unconnected sellers ($\beta^{C,U} = 0.193$), and connected buyers also do better than unconnected buyers because $\beta^{U,C} = -0.181$.
- But the real point is in the quartiles of z_0^{pe}. In the highest quartile (many peripheral sellers), the connectedness effect is much larger for sellers than for buyers; in the lowest quartile, the pattern flips. This is exactly the paper's crowded-side prediction.
- **Table II (D2C prices).** On pooled days, clients get better prices from connected dealers than from core dealers ($\beta^{cu,C} = 0.034$, $\beta^{C,cu} = -0.029$), and worse prices from unconnected dealers ($\beta^{cu,U} = -0.159$, $\beta^{U,cu} = 0.176$).
- Again, the pooled average hides strong heterogeneity. In the extreme quartiles of z_0^{pe}, the sign and magnitude of customer-price advantages depend on whether the client is effectively on the crowded or uncrowded side of the market.
- Bottom line: the tables are a warning to empirical researchers. Inventory imbalance is not just a control variable; it is part of the mechanism.

6 Short summary

- **Method.** A theoretical OTC model with core dealers, connected peripheral dealers, and unconnected peripheral dealers, plus simulated regressions to illustrate empirical implications.
- **Key mechanism.** Dealer market power comes from connectedness and from being on the uncrowded side of the market.
- **Main prediction.** Connectedness matters for prices only when peripheral dealers' aggregate inventory is imbalanced.
- **Client-market implication.** Connected dealers quote tighter prices to clients because they can unwind inventory more favorably.

- **Welfare implication.** Decentralized interdealer bargaining is generally inefficient and raises end-investor trading costs.

7 Conclusion

7.1 Core conclusion

- The paper shows that inventory management and network structure are deeply intertwined in OTC markets.
- A dealer's connectedness to the core matters only when aggregate inventory imbalances make outside options valuable.
- These frictions affect both interdealer prices and the quotes ultimately offered to clients.

7.2 Economic intuition

- When a dealer is on the crowded side of the market, finding a counterparty is hard.
- If that dealer is connected, he can still escape to the core market; if not, he is stuck bargaining from a weak position.
- That is why the value of network centrality rises when inventory imbalances become more severe.

7.3 Takeaway

This paper is useful because it explains why "dealer centrality" does not have a single unconditional effect on OTC prices. The effect depends on the state of aggregate inventory in the network. For empirical work, that is the main lesson: **to understand price dispersion in OTC markets, one must jointly measure who is connected and who is crowded**.